

Battered on the battery front

FOLLOWING THE recall of nearly 10 million laptop batteries from around the world due to the risk of self-inflammation, senior executives from the Japanese electronic giant Sony apologised recently with a superficial, seated bow. Surprisingly, the corporate heads did not feel it necessary to stand up and explain the reasons behind the manufacturing flaw and the corrective measures they plan to take to eliminate a recurrence of such a big mishap. Sony has admitted to a quarterly loss due to the astounding cost of recalling these batteries. They will spend approximately \$250 million replacing them for their customers. This incident has once again raised questions on the improvements that have been made in battery technology and the focus or funds consumer technology companies dedicate to research and development in this area.

What really caused the Sony Li-ion batteries to heat and go up in flames is called 'thermal runaway'. It happens when elements inside a Li-ion battery heat up to the point where the internal reaction accelerates, creating even more heat, finally resulting in a flameout. Flammability and heat intolerance are long-standing problems that have been the bane of Li-ion batteries since their invention almost 30 years ago. And now, as devices turn smaller in size and richer in features, the situation only worsens. If obligated to produce more energy in less space, Li-ion batteries tend to die faster. This was discovered in the first shipment of iPods wherein the batteries wore out faster than the players themselves.

Li-ion, with its high power, fast recharge times, and steady voltage, is the best battery the consumer electronics industry has today. It powered the 50 million laptops, 800 million cell phones, and 80 million digital cameras sold in 2005. If this technology stagnates without a viable replacement, so will every kind of portable device, from iPods to cell phones. *And* this technology is fast approaching its limits. In the last 150 years, battery performance has improved only about eight-fold. If we are indeed seeing the peak of Li-ion performance, then gadget makers (and users alike) are in trouble.



Deepak Ajwani Executive Editor

"If we are indeed seeing the peak of Li-ion performance, then gadget makers (and users alike) are in trouble"

Yes, the hunt for a new technology is on, but nothing concrete has been achieved so far. Possible successors like fuel cells have been touted as "options", but design, implementation, and cost issues have prevented them from mass adoption. It seems we'll need more than just better batteries; we'll need to rethink the way all portable electronics are designed and made.

A handwritten signature in black ink, reading 'Deepak Ajwani'.

editor@thinkdigit.com